

MONITORING & EVALUATION PORTFOLIO PROJECT

ENERGY DEMAND FORECASTING AND KPI MONITORING REPORT

Prepared by

Alex Maseko

BSc Applied Statistics (Credit) | MEAL Certified

+265 884 479 667 | alexmaseko166@gmail.com

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Tools: Stata 17 | Data: Simulated CDS (Jan 2018 – Dec 2024) | Forecast horizon: Jan – Dec 2025

1. Executive Summary

This report presents a structured Monitoring and Evaluation (M&E) analysis of Malawi's electricity generation and demand data from January 2018 to December 2024, with a 12-month forecast for the period January to December 2025. The analysis was conducted in Stata 17 using a Seasonal Autoregressive Integrated Moving Average (SARIMA) model, consistent with best-practice statistical approaches for time series data in the energy sector.

The purpose of this project is to demonstrate competency Data Analytics and M&E including: forecasting energy generation bounded to demand; predicting energy demand in reference to operational KPIs; validating and cleaning data from divisional reports; structuring findings into required periodic reporting formats; and informing executive management through monthly and quarterly performance reports.

Key findings

- National energy demand grew from approximately 293 GWh/month in 2018 to 398 GWh/month by end of 2024, a compound annual growth rate (CAGR) of approximately 5.2%.
- The generation adequacy ratio declined from 0.94 in 2018 to 0.92 in 2024, indicating that generation growth is not keeping pace with rising demand.
- Unmet energy demand averaged 24.5 GWh/month in 2024, representing an energy access gap of approximately 6% of total demand.
- The SARIMA(1,1,1)(1,1,0)₁₂ model achieved a Mean Absolute Percentage Error (MAPE) of 2.8% on the 2024 holdout sample, confirming strong predictive accuracy.
- Demand is forecast to reach between 386 and 422 GWh/month in January 2025, rising to between 389 and 460 GWh/month by mid-2025, with mid-year seasonal peaks driven by dry-season load characteristics.

2. Introduction and Background

2.1 Context

Electricity Generation Company (Malawi) Limited (EGENCO) is the national electricity generation utility, operating four hydro power stations namely Nkula, Tedzani, Kapichira, and Wovwe as well as thermal power plants in Lilongwe, Mzuzu, and Mapanga, with a total installed capacity of 444.67 MW. As Malawi's economy grows and electrification rates increase, accurate forecasting of energy demand and systematic monitoring of generation KPIs are critical to infrastructure planning, load management, and evidence-based reporting to Executive Management and government bodies.

2.2 Objectives

- Load, validate, and create a Consistent Data Set (CDS) from monthly energy data (2018–2024).
- Compute operational KPIs: generation adequacy ratio, capacity utilisation, energy access gap, hydro dependency, supply reliability index, and system losses.
- Conduct exploratory data analysis including seasonal decomposition and correlation analysis.
- Test for stationarity using the Augmented Dickey-Fuller test.
- Fit and compare ARIMA and SARIMA models; select the best-performing model by AIC/BIC.
- Forecast monthly energy demand and generation for January to December 2025 with 95% confidence intervals.
- Evaluate forecast accuracy using MAE, RMSE, and MAPE on a 2024 holdout sample.
- Export structured KPI monitoring tables and forecast results for executive reporting.

2.3 Data Description

The dataset contains 84 monthly observations from January 2018 to December 2024, constructed as a representative Consistent Data Set (CDS) for Malawi's electricity sector. Variables include:

Variable	Description	Unit
generation_gwh	Total electricity generated	GWh/month
demand_gwh	Total electricity demanded	GWh/month
peak_demand_mw	Monthly peak demand	MW
capacity_mw	Total installed generation capacity	MW

Variable	Description	Unit
hydro_gwh	Generation from hydro sources	GWh/month
thermal_gwh	Generation from thermal sources	GWh/month
load_factor_pct	Load factor	Percent
unmet_demand_gwh	Demand not met by generation	GWh/month
system_losses_pct	Transmission and distribution losses	Percent
rainfall_mm	Monthly rainfall (proxy for hydro potential)	mm
gdp_index	Economic activity index (2018=100)	Index

3. Methodology

3.1 Data Validation and Cleaning

Following the principles of Monitoring and Evaluation data management, the raw dataset was subjected to systematic validation before analysis:

- Completeness check: the misstable command in Stata confirmed zero missing values across all 84 observations.
- Range validation: all generation and demand values were verified to fall within physically plausible bounds (0 to capacity equivalent).
- Consistency check: $\text{hydro_gwh} + \text{thermal_gwh}$ was verified to equal generation_gwh within rounding tolerance for all observations.
- Time series integrity: the dataset was declared as monthly time series using `tsset`, and the panel was confirmed as strongly balanced with no gaps.

3.2 KPI Computation

Six operational KPIs were derived from the validated dataset:

- Generation adequacy ratio = $\text{generation_gwh} / \text{demand_gwh}$. A ratio below 1.0 indicates a supply shortfall.
- Capacity utilisation (%) = $(\text{generation_gwh} / (\text{capacity_mw} \times 24 \times 30 / 1000)) \times 100$. Measures how effectively installed capacity is being used.
- Energy access gap (%) = $(\text{unmet_demand_gwh} / \text{demand_gwh}) \times 100$. Measures the proportion of demand that cannot be met.
- Hydro dependency ratio (%) = $(\text{hydro_gwh} / \text{generation_gwh}) \times 100$. Tracks exposure to hydrological risk.
- Supply reliability index = $100 - \text{access_gap_pct}$. Higher values indicate greater system reliability.
- Year-on-year demand growth (%) = $(\text{demand_t} - \text{demand_t-12}) / \text{demand_t-12} \times 100$.

3.3 Time Series Analysis and Model Selection

Prior to model fitting, the Augmented Dickey-Fuller (ADF) test was applied to test for a unit root in the demand series. The null hypothesis of non-stationarity was not rejected at the 5% level for the level series ($p > 0.05$), but was rejected for the first difference ($p < 0.01$), confirming $d = 1$ (one non-seasonal difference required).

Visual inspection of the autocorrelation function (ACF) and partial autocorrelation function (PACF) indicated significant spikes at lag 12, confirming annual seasonality consistent with Malawi's rainfall-driven hydro dependency and seasonal load patterns. Accordingly, four candidate models were evaluated:

Model	Specification	AIC	BIC	Notes
A	ARIMA(1,1,1)	487.3	496.1	Baseline; misses seasonality
B	ARIMA(2,1,1)	485.9	497.2	Marginal improvement
C	SARIMA(1,1,1)(1,1,0)12	461.2	473.8	Best AIC; captures annual cycle
D	SARIMA(1,1,1)(0,1,1)12	463.7	476.3	Close; slightly higher AIC

Model C — SARIMA(1,1,1)(1,1,0)12 — was selected as the preferred model based on the lowest AIC (461.2) and sound residual diagnostics. The model was estimated on a training window of January 2018 to December 2023 (72 observations), with January to December 2024 reserved as the holdout test window (12 observations).

3.4 Forecast Evaluation

Forecast accuracy on the 2024 holdout was assessed using three standard metrics:

- Mean Absolute Error (MAE): average absolute deviation between forecast and actual demand.
- Root Mean Squared Error (RMSE): penalises large errors more heavily than MAE.
- Mean Absolute Percentage Error (MAPE): scale-independent accuracy measure expressed as a percentage.

The model was then refit on the full 84-month dataset before generating the final 12-month forecast for January to December 2025.

4. Results

4.1 KPI Monitoring Summary — Annual Averages

Table 2 presents the annual KPI monitoring summary for the period 2018–2024.

Year	Gen (GWh)	Demand (GWh)	Adequacy Ratio	Cap Util %	Access Gap %	Reliability Idx	Losses %
2018	277.8	295.2	0.94	22.4	17.7	88.3	82.2
2019	293.1	312.6	0.94	22.7	17.3	87.9	81.8
2020	305.4	327.8	0.93	23.5	16.8	87.1	80.2
2021	319.2	344.1	0.93	24.1	16.3	86.4	79.6
2022	333.8	360.9	0.93	24.8	16.0	85.8	78.9
2023	350.1	379.2	0.92	25.2	15.5	85.1	78.1
2024	366.4	398.7	0.92	25.9	15.2	84.3	77.0

Key observations from the KPI monitoring table:

- Generation adequacy declined consistently from 0.94 in 2018 to 0.92 in 2024, signalling growing pressure on the generation fleet as demand outpaces supply.
- Capacity utilisation fell from 22.4% to 25.9%, reflecting the gap between installed capacity (444 MW) and actual dispatch, largely due to hydrological constraints on hydro plants during dry seasons.
- The energy access gap widened from 6.7% (2018) to 8.1% (2024), representing an increase in unmet demand from approximately 19 GWh/month to 32 GWh/month.
- System losses declined from 17.7% to 15.2%, suggesting improving transmission and distribution efficiency.

4.2 Seasonal Decomposition

Analysis of the monthly seasonal pattern revealed that energy demand peaks in the dry season months of June to September consistent with reduced ambient temperatures increasing heating demand and reduced surface water availability increasing groundwater pumping loads. Hydro generation is inversely seasonal, peaking in January to April following the rainy season and declining sharply in July to October.

The negative correlation between hydro generation and demand seasonality is a structural vulnerability in Malawi's power system: demand is highest precisely when hydro output is lowest.

This reinforces the importance of accurate demand forecasting to plan for thermal backup dispatch and demand management interventions during peak deficit periods.

4.3 Rainfall–Hydro Generation Relationship

Regression analysis confirmed a significant positive relationship between monthly rainfall and hydro generation ($R^2 = 0.58$, $p < 0.001$), validating rainfall as a useful leading indicator for hydro output projections in the monitoring framework. A one-millimetre increase in monthly rainfall is associated with approximately 0.7 GWh of additional hydro generation on average.

4.4 Forecast Accuracy — 2024 Holdout

Metric	Value	Interpretation
MAE	8.4 GWh	Average monthly forecast error
RMSE	10.7 GWh	RMSE slightly higher — a few months with larger errors
MAPE	2.8%	Strong accuracy; industry threshold typically $\leq 5\%$

A MAPE of 2.8% confirms that the SARIMA(1,1,1)(1,1,0)₁₂ model is well-specified and suitable for operational demand forecasting. The small gap between MAE and RMSE indicates no extreme outlier months in the holdout period.

4.5 Demand Forecast — January to December 2025

Table 4 presents the monthly demand forecast for 2025 with 95% confidence intervals, structured for direct use in EGENCO's monthly reporting cycle.

Month	Forecast (GWh)	Lower 95% CI	Upper 95% CI	Std Error
Jan 2025	386.4	350.8	422.0	18.1
Feb 2025	374.2	339.1	409.3	17.6
Mar 2025	371.8	336.8	406.8	17.7
Apr 2025	390.5	354.9	426.1	18.2
May 2025	402.1	365.8	438.4	18.5
Jun 2025	415.3	378.4	452.2	18.9
Jul 2025	421.7	384.2	459.2	19.1
Aug 2025	418.6	381.3	455.9	18.9
Sep 2025	408.3	371.5	445.1	18.6
Oct 2025	395.7	359.4	432.0	18.3

Month	Forecast (GWh)	Lower 95% CI	Upper 95% CI	Std Error
Nov 2025	389.4	353.4	425.4	18.2
Dec 2025	393.2	356.9	429.5	18.3

The forecast indicates continued demand growth in 2025, with peak demand expected in the June to August dry season window (415–422 GWh/month). The confidence intervals widen over the forecast horizon, reflecting increasing uncertainty, a standard property of time series forecasts. Monitoring officers should trigger a review if actual monthly demand exceeds the upper confidence bound for two or more consecutive months, as this may signal an accelerating growth trend requiring revised generation planning.

5. Conclusions and M&E Recommendations

5.1 Conclusions

- The SARIMA(1,1,1)(1,1,0)12 model provides accurate and reliable monthly demand forecasts for operational planning, with a MAPE of 2.8% on holdout data.
- Malawi's energy demand is growing at approximately 5.2% per annum, driven by economic growth (GDP index) and expanding access, while generation capacity has remained constrained at 444 MW.
- The widening energy access gap (6.7% in 2018 to 8.1% in 2024) represents a persistent deficit that worsens during dry-season months. Without additional generation capacity, this gap is projected to reach approximately 9–10% by 2026.
- Hydro generation is highly sensitive to rainfall variability, creating a structural seasonal mismatch between supply and demand that the monitoring framework must explicitly track.

5.2 M&E Recommendations

- Implement a monthly CDS update cycle: collect divisional generation and demand actuals by the 5th of each month, validate against prior-month trends, and flag anomalies exceeding ± 2 standard deviations.
- Establish quarterly forecast reviews: re-estimate the SARIMA model each quarter using updated actuals to recalibrate the 12-month rolling forecast.
- Develop a monitoring trigger system: flag months where actual demand exceeds the upper 95% CI forecast as requiring immediate review by the M&E Manager and relevant Divisions.
- Integrate rainfall monitoring data: incorporate actual rainfall observations from the Department of Climate Change and Meteorological Services as a leading indicator in the monitoring dashboard.
- Align with EGENCO's programme theory of change: link the demand forecasting KPIs explicitly to EGENCO's strategic objective of reliable power supply, so that monitoring reports communicate not just what happened, but whether the programme is on track to meet national energy access targets.

6. Technical Appendix

6.1 Software and Files

File	Description
malawi_energy_data.csv	Raw CDS input — 84 monthly observations, 14 variables
energy_forecasting.do	Complete Stata do-file: data import → KPI computation → EDA → ARIMA/SARIMA → forecast → export
malawi_energy_clean.dta	Validated Stata dataset with computed KPIs
kpi_annual_summary.csv	Annual KPI monitoring table (exported from Stata)
forecast_2025_monthly.csv	Monthly demand forecast with CIs (exported from Stata)
energy_forecasting_log.smcl	Full Stata log of all results
fig1_demand_vs_generation.png	Time series: demand vs generation
fig2_unmet_demand.png	Unmet demand trend
fig3_seasonal_pattern.png	Average monthly seasonal pattern
fig4_rainfall_hydro.png	Rainfall–hydro scatter with linear fit
fig5_acf_demand.png	ACF — demand levels
fig6_pacf_demand.png	PACF — demand levels
fig7_acf_diff_demand.png	ACF — first difference
fig8_forecast_vs_actual.png	SARIMA forecast vs actual (2024 holdout)
fig9_demand_forecast_2025.png	2025 demand forecast with 95% CI
fig10_gen_demand_forecast_2025.png	2025 generation vs demand forecast

6.2 SARIMA Model Specification

The selected model SARIMA(1,1,1)(1,1,0)₁₂ is specified as:

$$\Phi(B^s) \varphi(B) \nabla^s \nabla Y_t = \theta(B) \varepsilon_t$$

Where: $d=1$ (one non-seasonal difference), $D=1$ (one seasonal difference at lag $s=12$), AR order $p=1$, MA order $q=1$, seasonal AR order $P=1$. The model was estimated by maximum likelihood. Residual diagnostics confirmed white-noise residuals with no remaining autocorrelation (Ljung-Box Q statistic $p > 0.10$ at all lags tested).

6.3 GitHub Repository Structure

This project is available at: github.com/alex166-cmd/energy-forecasting-malawi

Folder / File	Contents
README.md	Project overview, methodology summary, how to run the do-file
data/malawi_energy_data.csv	Input dataset
stata/energy_forecasting.do	Main do-file — run this to reproduce all results
outputs/kpi_annual_summary.csv	Annual KPI table
outputs/forecast_2025_monthly.csv	2025 forecast table
figures/	All 10 exported charts
report/Malawi_Energy_Forecasting_Report.docx	This report

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